

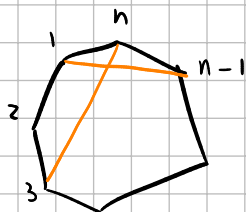
if we have an amplitude with only 1 chord and n vertices, we can show that if the amplitude satisfies $\neq$ distinct 1-zero then everything vanishes.

$$B = \sum_{(i,j)} \frac{a_{ij}}{X_{ij}}$$

1-zero $\neq 1$ $X_{2j} = X_{ij} - X_{13} \quad j = 4, \dots, n-1$
 $X_{2n} = -X_{13}$

$\rightarrow$ only survivors are $\frac{a_{13}}{X_{13}}$ and $\frac{a_{2n}}{X_{2n}}$

for another 1-zero at least one of 13 and 2n is not bare
 so it's killed and since $B=0$ the other one is also killed



next to look Res X_{1n-1}, X_{2n}

project down as many zeros as possible

what's left is an amplitude for $n=6$ with 2 chord only

the two remaining don't go to n

$$B = \sum_{(i_1, j_1) < (i_2, j_2)} \frac{a_{i_1 j_1} a_{i_2 j_2}}{X_{i_1 j_1} X_{i_2 j_2}}$$

$(i_1, j_1) < (i_2, j_2) \Leftrightarrow i_1 < i_2$ or $i_1 = i_2$ and $j_1 < j_2$

can we make them all the propagators in the denominator born free

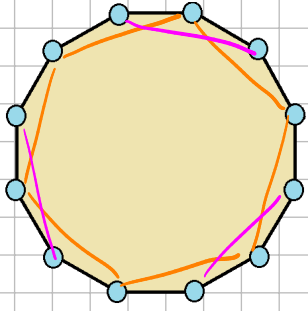
try $n=6$

(BF)

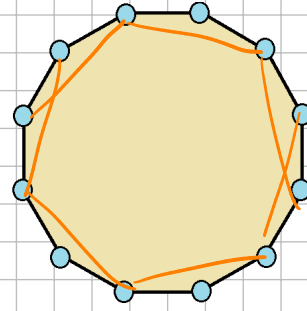
$$\begin{array}{c|c|c|c|c|c} 35 & 36 & 46 & 41 & 51 & 52 \\ \hline & 46 & & 51 & & 62 \\ \hline 13 & 46 & 24 & 51 & 35 & 62 \end{array}$$

$$\begin{array}{c|c|c|c|c} 13 & 14 & 15 & 26 & 25 \\ \hline & 24 & 25 & & 36 \\ \hline & & 35 & 36 & 46 \end{array}$$

And make all orientations of 2 propagation born free for some zero



$$\lfloor \frac{n}{2} \rfloor + \lfloor \frac{n}{4} \rfloor$$



aside:

